



Personal safety wearable that works with a phone or on its own. Sensors and on-device AI detect risk locally. Alerts go out over Bluetooth to the app, or over cellular with an eSIM when the phone is not there. The path is wearable → edge AI → BLE / cellular → mobile → cloud → secure alerts.

THE GAP

Most safety products are a button, a location share, or a wearable that must stay paired to a phone. They fail when the phone is away, Bluetooth drops, the user cannot press a button, raw sensor data is not interpreted, or personal and location data is not protected. SentriX Core keeps sensing, detection, secure communication, and response in one system.

FIVE LAYERS

01

Wearable

MCU, sensors, battery, and embedded software.

02

Edge AI

On-device inference. Not all raw data goes to the cloud.

03

Connectivity

BLE with the phone, or cellular modem plus eSIM alone.

04

Mobile & cloud

App, location, notifications, APIs, storage, and monitoring.

05

Security

Auth, encryption, access control, and API protection.

WITH SMARTPHONE

Wearable → BLE → app → internet → backend

STANDALONE

Wearable → cellular / eSIM → backend

WHO IT IS FOR

Children

Monitoring, location, and alerts when their phone is unavailable.

Older adults

Continuous monitoring and alerts from the wearable itself.

Closer monitoring

People who need continuous watch and fast notification.

General use

Commuting, travel, and outdoor situations.

WHERE IT STANDS

AI workflow	~90%
Mobile app	~50%
Security plan	Done

AI is through training, evaluation, the risk pipeline, and deployment prep. The app has architecture, core UI, and the communication plan. Hardware, on-device deployment, and full-path integration are still ahead.

COMPONENT	STATUS
AI workflow	~90%
AI deployment & validation	In progress
Mobile application	~50%
Security architecture	Completed
Security implementation	In progress
Backend / cloud	In development
Hardware	Prototyping
Cellular / eSIM	Planned
System integration	In progress
End-to-end & physical tests	Pending

SCOPE

DOMAIN

WHAT IT COVERS

AI / ML & edge AI	Risk, prediction, anomaly detection, on-device inference, quantization, TinyML / TensorFlow Lite Micro
Hardware	MCU, sensors, PCB, battery, power, enclosure
Connectivity	BLE, cellular modem, eSIM, provisioning, low-power cellular
Mobile & cloud	App, location, notifications, APIs, storage, monitoring
Cybersecurity	Device identity, authentication, encryption, API security, privacy

BUSINESS & FUNDING

A hardware-plus-service path, funded to reach a validated prototype

Business model

Revenue can come from selling the wearable, plus optional recurring services for connectivity, cloud monitoring, advanced alerts, and analytics. The same platform can be offered to institutions through partnerships with organizations that need connected safety.

Funding objective

Funding moves the existing AI and software into one integrated, validated prototype. It goes to hardware prototyping, PCB fabrication, cellular/eSIM integration, embedded AI deployment, system integration, testing, and preparation for a first pilot.

3-year commercial projection — illustrative

YEAR	DEVICES	AVERAGE REVENUE / DEVICE*	PROJECTED REVENUE*
Year 1	500	EGP 2,500	EGP 1.25M
Year 2	2,000	EGP 2,300	EGP 4.60M
Year 3	5,000	EGP 2,100	EGP 10.50M

* Illustrative planning assumptions for discussion. Final figures follow prototype validation, pricing, market testing, and commercialization planning.

Commercialization path

Prototype **Validation** **Pilot** **Commercialization**

The integrated prototype is validated, then tried in a controlled pilot. Pilot results drive product changes, manufacturing preparation, pricing, and later commercial partnerships.

WAYS TO COLLABORATE

Support can take more than one form

Technical mentorship

Guidance from people in embedded systems, IoT, AI/ML, cybersecurity, mobile, cloud, and hardware engineering.

Hardware sponsorship

Components, development boards, sensors, cellular modules, PCBs, batteries, and other materials for the prototype.

Financial sponsorship

Support for procurement, PCB fabrication, prototype assembly, testing and validation, and development resources.

Prototyping & testing

Access to laboratories, equipment, testing facilities, or technical resources where they are available.

Engineering review

Professional feedback on hardware architecture, embedded implementation, AI deployment, security architecture, cellular connectivity, and system integration.

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